Lecture 17

Curvature

Objectives. *Second covariant derivatives need not commute. Their difference, corrected for the bracket of the directions, defines curvature. We compute this tensor from Christoffel symbols and from differential forms in an orthonormal frame. The sphere fixes the sign used for surface geometry and gravitation.*

By the end of this lecture you will be able to:

- Compute curvature with the output and direction indices specified.
- Prove tensoriality, the metric antisymmetries and the first Bianchi identity.
- Use Cartan's equations to find the curvature of a surface.
- Distinguish zero curvature from zero Christoffel symbols in one chart.

17.1 The curvature tensor

Let M be a smooth manifold with an affine connection ∇. The *curvature operator* is defined on smooth fields by

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

For the Cartesian connection on $\mathbb{R}^n$, its components are $XY(Z^k) - YX(Z^k) - [X,Y](Z^k) = 0$. The bracket correction removes the failure of the direction fields to commute.

Theorem 17.1 (Tensoriality of curvature)**.** *Curvature is linear over $C^\infty(M)$ in all three arguments. It therefore gives a smooth tensor of type $(1,3)$, called the* Riemann curvature tensor.

Proof. Real multilinearity is immediate from the connection rules. For $f \in C^\infty(M)$, the Leibniz rule and $[fX,Y] = f[X,Y] - Y(f)X$ give

$$R(fX,Y)Z = fR(X,Y)Z - Y(f)\nabla_X Z + Y(f)\nabla_X Z = fR(X,Y)Z.$$

Antisymmetry in X and Y gives linearity in Y. Expansion in the last slot gives

$$R(X,Y)(fZ) = fR(X,Y)Z + \big(XY(f) - YX(f) - [X,Y](f)\big)Z = fR(X,Y)Z.$$

The cross terms $Y(f)\nabla_X Z$ and $X(f)\nabla_Y Z$ cancel between the two second derivatives. By the cutoff and coordinate-expansion argument of § 16.1, the value therefore depends only on the three vectors at the point. Smoothness follows from the local coefficients of the connection. ∎

Write $\partial_\mu = \partial/\partial x^\mu$ in a chart. Greek indices here range from 1 to n. The index order is fixed by the action

$$R(\partial_\mu, \partial_\nu)\partial_\sigma = R^\rho{}_{\sigma\mu\nu}\partial_\rho.$$

Applying the connection product rule twice gives

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}.$$

For example, applying the product rule gives

$$\nabla_{\partial_\mu}\nabla_{\partial_\nu}\partial_\sigma = (\partial_\mu \Gamma^{\rho}_{\nu\sigma} + \Gamma^{\lambda}_{\nu\sigma}\Gamma^{\rho}_{\mu\lambda})\partial_\rho.$$

Subtract the reversed expression. The bracket term vanishes because coordinate fields commute. Curvature is a tensor even though the Christoffel symbols are not. For the Euclidean plane, Lecture 16 gives the polar symbols $\Gamma^{r}_{\phi\phi} = -r$ and $\Gamma^{\phi}_{r\phi} = \Gamma^{\phi}_{\phi r} = 1/r$. For example, the curvature component is

$$R^{r}{}_{\phi r\phi} = \partial_r(-r) - (-r)(1/r) = -1 + 1 = 0.$$

All curvature components vanish because the tensor is zero in Cartesian coordinates. A connection is *flat* when its curvature is zero. On the unit sphere, the calculation in Example 17.6 below gives $R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta > 0$ in the spherical chart of Lecture 16. This is the sphere sign test for our operator convention.

17.2 Metric symmetries and the first Bianchi identity

Now let ∇ be the Levi-Civita connection of a pseudo-Riemannian metric. Lower the output index first: $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^{\lambda}{}_{\sigma\mu\nu}$. For the proofs, abbreviate the four-field expression by $C(X,Y,Z,W) = g(R(X,Y)Z,W)$.

Proposition 17.2 (Curvature symmetries · *sketched; antisymmetries and first Bianchi proved*)**.** *The curvature satisfies*

$$\begin{aligned} C(X,Y,Z,W) &= -C(Y,X,Z,W) = -C(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0, \\ C(X,Y,Z,W) &= C(Z,W,X,Y). \end{aligned}$$

The middle equation is the first Bianchi identity.

Proof and sketch. The first antisymmetry follows from the operator definition. For the other one, apply $XY - YX - [X,Y]$ to $g(Z,W)$. It vanishes on this scalar function. Metric compatibility expands the result as

$$0 = g(R(X,Y)Z,W) + g(Z,R(X,Y)W),$$

with the two cross products of first derivatives cancelling. This proves antisymmetry in the last pair.

For Bianchi, sum cyclically over (X,Y,Z). Zero torsion first combines the second derivatives into $\sum_{\text{cyc}} \nabla_X[Y,Z]$. Relabel the bracket-direction terms to obtain

$$\sum_{\text{cyc}} R(X,Y)Z = \sum_{\text{cyc}} \big(\nabla_X[Y,Z] - \nabla_{[Y,Z]}X\big) = \sum_{\text{cyc}} [X,[Y,Z]] = 0.$$

The last equality is Jacobi's identity.

For pair exchange, write Bianchi for four cyclic permutations of the four fields. Add the identities and use the two antisymmetries to cancel terms. The remaining equality gives pair exchange after renaming the fields. This sketches the algebraic step; see Lee, Proposition 7.4, pp. 121–123. ∎

In the lowered-output convention, these identities give $R_{abcd} = -R_{bacd} = -R_{abdc}$ and $R_{abcd} = R_{cdab}$. The slot order must also be translated when comparing sources.[1] Changing the operator sign would negate every curvature value below; changing slot order alone can also introduce a sign.

17.3 Ricci, scalar and sectional curvature

The *Ricci tensor* is the trace $\mathrm{Ric}(X,Y) = \mathrm{tr}(Z \mapsto R(Z,Y)X)$. The *scalar curvature* is its metric trace. In coordinates,

$$\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}\,.$$

Pair exchange makes Ricci symmetric: in $\mathrm{Ric}_{ij} = g^{ab}R_{aibj}$, exchange the curvature pairs and then rename a and b. The symbol R_{sc} distinguishes scalar curvature from a sphere radius.

For a Riemannian metric and a tangent 2-plane $P \subset T_pM$, its *sectional curvature* is

$$K(P) = \frac{g(R(u,v)v,u)}{g(u,u)g(v,v) - g(u,v)^2}, \qquad P = \mathrm{span}\{u,v\}.$$

Here u and v are linearly independent. The denominator is positive. On changing the basis of P, antisymmetry in both pairs multiplies the numerator by the square of the basis-change determinant. The Gram determinant in the denominator has the same factor. Thus the quotient is independent of the basis. For orthonormal u and v, it is $g(R(u,v)v,u)$. In indefinite signature the same quotient defines sectional curvature only for nondegenerate planes, whose Gram determinant is nonzero.

17.4 Moving frames and Cartan's equations

A *local frame* $e = (e_1,\dots,e_n)$ is a smooth basis of tangent vectors at every point of an open set U. It is a row of fields. Its *dual coframe* is the column of 1-forms $\theta = (\theta^1,\dots,\theta^n)^T$ with $\theta^i(e_j) = \delta^i_j$. For a Riemannian metric, a frame is *orthonormal* when $g(e_i,e_j) = \delta_{ij}$. Gram–Schmidt applied smoothly to a coordinate frame supplies local orthonormal frames. The construction divides only by positive smooth lengths. The dual coframe then gives $g = \sum_i \theta^i \otimes \theta^i$. The figure compares coordinate vectors with unit frame vectors on a sphere.

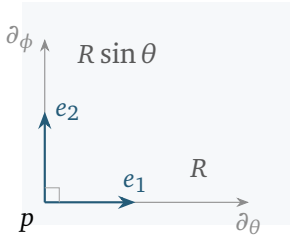


Figure 17.1: For a radius-R sphere, the spherical coordinate vectors have lengths R and $R\sin\theta$. Dividing by those lengths gives the orthonormal frame e_1 and e_2 on a chart away from the poles.

The *connection forms* $\omega^i{}_j$ and *curvature forms* $\Omega^i{}_j$ are specified by

$$\nabla_X e_j = e_i\omega^i{}_j(X), \qquad R(X,Y)e_j = e_i\Omega^i{}_j(X,Y).$$

[1] Lee uses $\mathrm{Rm}_{\mathrm{Lee}}(X,Y,Z,W) = C(X,Y,Z,W)$. Thus $R_{abcd} = \mathrm{Rm}_{\mathrm{Lee}}(\partial_c,\partial_d,\partial_b,\partial_a)$, with the same operator sign.

Linearity of the connection in its direction makes the first coefficients 1-forms. Tensoriality and direction antisymmetry make the second 2-forms. Write ∇V for the vector-valued 1-form $X \mapsto \nabla_X V$. The matrices act on coefficient columns: for $V = e v$, $\nabla V = e(\mathrm{d}v + \omega v)$. For matrices of forms, the wedge includes matrix multiplication: $(A \wedge B)^i{}_j = \sum_k A^i{}_k \wedge B^k{}_j$. The exterior derivative acts on each entry.

Theorem 17.3 (Cartan's structure equations)**.** *On a Riemannian manifold, the Levi-Civita connection in a local orthonormal frame satisfies*

$$\mathrm{d}\theta = -\omega \wedge \theta, \qquad \Omega = \mathrm{d}\omega + \omega \wedge \omega.$$

The connection matrix ω and curvature matrix Ω are skew-symmetric. Conversely, a skew matrix ω satisfying the first equation is the unique Levi-Civita connection matrix for this coframe.

Proof. Differentiate $g(e_i, e_j) = \delta_{ij}$. Metric compatibility gives $\omega^i{}_j + \omega^j{}_i = 0$. For any 1-form α, the expression $X\alpha(Y) - Y\alpha(X) - \alpha([X,Y])$ is function-linear in X and Y by the product and bracket rules. On coordinate fields it agrees with Lecture 10's definition of $\mathrm{d}\alpha$, so the expressions coincide. For any connection matrix in this frame, the same evaluation gives

$$\theta^i(T(X,Y)) = (\mathrm{d}\theta^i + \omega^i{}_j \wedge \theta^j)(X,Y).$$

Indeed, expand $Y = \theta^j(Y) e_j$ in $\nabla_X Y$ and subtract the reversed expression and $[X,Y]$. Zero torsion therefore gives the first structure equation.

Conversely, a smooth matrix ω defines a connection by $\nabla_X(ev) = e(Xv + \omega(X)v)$. The product rule verifies both connection axioms. If ω is skew, the extra terms in differentiating $u^T v$ cancel because $u^T(\omega(X)^T + \omega(X))v = 0$. Thus this connection is metric. The displayed torsion identity makes it torsion-free when the first equation holds. Uniqueness in Theorem 16.1 identifies it with the Levi-Civita connection, proving the converse and uniqueness of the skew solution.

For the second, expand $\nabla_X \nabla_Y e_j$ using the Leibniz rule and subtract its reversed expression and $\nabla_{[X,Y]} e_j$. The coefficient of e_i is

$$X\omega^i{}_j(Y) - Y\omega^i{}_j(X) - \omega^i{}_j([X,Y]) + \omega^i{}_k(X)\omega^k{}_j(Y) - \omega^i{}_k(Y)\omega^k{}_j(X).$$

The first terms are $\mathrm{d}\omega^i{}_j(X,Y)$ and the last terms are $(\omega \wedge \omega)^i{}_j(X,Y)$. This proves the curvature equation. The last-pair metric antisymmetry already proved gives $\Omega^i{}_j = -\Omega^j{}_i$. ∎

The curvature-form calculation used no metric hypothesis. Thus $\Omega = \mathrm{d}\omega + \omega \wedge \omega$ holds in any local frame for any affine connection. Differentiate a tensor by the connection product rule: $\nabla_X T$ differentiates the output and subtracts a term for the covariant derivative of each input. For curvature, this means

$$\begin{aligned}(\nabla_X R)(Y,Z)W = {} & \nabla_X(R(Y,Z)W) - R(\nabla_X Y, Z)W \\ & - R(Y, \nabla_X Z)W - R(Y,Z)\nabla_X W.\end{aligned}$$

Proposition 17.4 (Second Bianchi identity · *sketched; matrix-form identity proved*)**.** *For the Levi-Civita connection of a pseudo-Riemannian metric,*

$$\mathrm{d}\Omega + \omega \wedge \Omega - \Omega \wedge \omega = 0, \qquad \sum_{\mathrm{cyc}(X,Y,Z)} (\nabla_X R)(Y,Z) = 0.$$

Proof and sketch. The identity $\mathrm{d}^2\omega = 0$ gives $\mathrm{d}\Omega = \mathrm{d}\omega\wedge\omega - \omega\wedge\mathrm{d}\omega$. Substitute $\mathrm{d}\omega = \Omega - \omega\wedge\omega$; the cubic terms cancel. This proves the matrix equation. Evaluating it on three directions and using zero torsion cancels their bracket terms, giving the stated cyclic covariant identity. This sketches the conversion from forms to the covariant expression. Nakahara's (7.147b) states the matrix identity with the same action convention. ∎

17.5 Gaussian curvature of surfaces

On a Riemannian surface, the *Gaussian curvature* K is the sectional curvature of its tangent plane. Choose a positively oriented orthonormal frame on a chart. Write its connection entry as $\omega^1{}_2$. Skew-symmetry makes the matrix

$$\omega = \begin{pmatrix} 0 & \omega^1{}_2 \\ -\omega^1{}_2 & 0 \end{pmatrix}.$$

Cartan's equations become

$$\mathrm{d}\theta^1 = -\omega^1{}_2 \wedge \theta^2, \qquad \mathrm{d}\theta^2 = \omega^1{}_2 \wedge \theta^1, \qquad \mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2.$$

For the last equality, the diagonal-free 2×2 matrix has $(\omega\wedge\omega)^1{}_2 = 0$. The coefficient of $\Omega^1{}_2$ on (e_1, e_2) is $g(R(e_1,e_2)e_2, e_1) = K$. Thus the curvature coefficient has the displayed positive sign. In terms of the frame, the same convention is $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$.

Proposition 17.5 (Surface curvature)**.** *On a Riemannian surface, the lowered-output curvature satisfies*

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}), \qquad \mathrm{Ric} = Kg, \qquad R_{\mathrm{sc}} = 2K.$$

The round sphere of radius $R > 0$ has $K = 1/R^2$. The Poincaré half-plane with metric $(\mathrm{d}x\otimes\mathrm{d}x + \mathrm{d}y\otimes\mathrm{d}y)/y^2$ has $K = -1$. The torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ with its quotient Euclidean metric has $K = 0$.

In two dimensions, antisymmetry in both pairs leaves a single independent curvature component in an orthonormal basis. Its value is K. Tensoriality therefore gives the displayed formula in any coordinates. Contracting its first and third indices with the inverse metric gives $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. This is intrinsic; no embedding is required. Tu's Theorem 12.3, p. 90, identifies it with classical surface curvature.[2]

The sphere and half-plane calculations follow in the worked example. For the torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$, use the metric induced by the Euclidean plane under the quotient charts. Their transition maps are translations, so the metric agrees on overlaps. Its coordinate coefficients are constant, hence its Christoffel symbols and curvature vanish. The figure shows why the Cartesian frame also descends.

[2] Read Tu's entry by its action on the frame: $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$. His (12.9) then reads $\mathrm{d}\omega^1{}_2 = +K\theta^1\wedge\theta^2$.

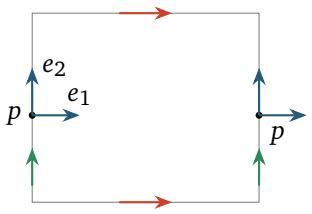


Figure 17.2: Opposite sides are identified by translations. The same Cartesian orthonormal frame matches across both pairs, giving the flat quotient metric.

This metric differs from the metric induced on a doughnut-shaped surface in $\mathbb{R}^3$.

17.6 The sphere by coordinates and by forms

Example 17.6 (Two curvature calculations)**.** On the sphere of radius $R > 0$, spherical coordinates give

$$g = R^2 \mathrm{d}\theta \otimes \mathrm{d}\theta + R^2 \sin^2\theta \mathrm{d}\phi \otimes \mathrm{d}\phi, \qquad 0 < \theta < \pi.$$

The constant scale does not change Lecture 16's symbols. Substitute them into the coordinate formula:

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= \partial_\theta(-\sin\theta\cos\theta) - (-\sin\theta\cos\theta)\cot\theta \\ &= (\sin^2\theta - \cos^2\theta) + \cos^2\theta = \sin^2\theta. \end{aligned}$$

Lowering the output gives $R_{\theta\phi\theta\phi} = R^2\sin^2\theta$. Divide by the Gram determinant $R^4\sin^2\theta$ to obtain $K = 1/R^2$. In particular the unit sphere has positive curvature, fixing the operator sign.

For the forms calculation use

$$\theta^1 = R\mathrm{d}\theta, \qquad \theta^2 = R\sin\theta\mathrm{d}\phi.$$

Their derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = R\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$. The first structure equation has the unique skew solution

$$\omega^1{}_2 = -\cos\theta\mathrm{d}\phi,$$

since $-\omega^1{}_2 \wedge \theta^2 = 0$ and $\omega^1{}_2 \wedge \theta^1 = R\cos\theta\mathrm{d}\theta \wedge \mathrm{d}\phi$. It follows that

$$\mathrm{d}\omega^1{}_2 = \sin\theta\mathrm{d}\theta \wedge \mathrm{d}\phi = \frac{1}{R^2}\theta^1 \wedge \theta^2.$$

Uniqueness of the Levi-Civita connection makes this the same curvature. Both routes give $\mathrm{Ric} = g/R^2$ and $R_{\mathrm{sc}} = 2/R^2$. These are smooth tensor identities on a dense coordinate domain, so continuity extends them, and $K = 1/R^2$, across the poles and angular seams.

On the half-plane $y > 0$, choose instead $\theta^1 = \mathrm{d}x/y$ and $\theta^2 = \mathrm{d}y/y$. The derivatives and solution of the first equation are

$$\mathrm{d}\theta^1 = \frac{\mathrm{d}x \wedge \mathrm{d}y}{y^2}, \qquad \mathrm{d}\theta^2 = 0, \qquad \omega^1{}_2 = -\frac{\mathrm{d}x}{y}.$$

Taking its derivative gives $\mathrm{d}\omega^1{}_2 = -\mathrm{d}x \wedge \mathrm{d}y/y^2 = -\theta^1 \wedge \theta^2$, so $K = -1$.

Self-check. Compute the intrinsic Gaussian curvature of the cylinder of radius $a > 0$.

Answer. The parametrization $p(u, z) = (a\cos u, a\sin u, z)$ induces $g = a^2 \mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z$. Use $\theta^1 = a\mathrm{d}u$ and $\theta^2 = \mathrm{d}z$ on an angular chart. Both forms are closed, so the skew connection matrix $\omega = 0$ satisfies Cartan's first equation and is the matrix of the unique metric torsion-free connection. Its curvature is zero, hence $K = 0$. The cylinder bends in its embedding but has zero intrinsic curvature.

Summary. Curvature is the tensor $R(X, Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}$. Metric compatibility and zero torsion give its antisymmetries and first Bianchi identity. Ricci and scalar curvature are its specified contractions. In a local orthonormal frame, Cartan's equations replace Christoffel computations by $\mathrm{d}\theta = -\omega \wedge \theta$ and $\Omega = \mathrm{d}\omega + \omega \wedge \omega$. On a surface, $\mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2$. The sphere gives positive curvature, the half-plane negative curvature, and the flat quotient torus and cylinder zero curvature. Lecture 18 integrates the surface equation to compute parallel transport around loops.

Reading. Tu, *Differential Geometry*, §10–§12, especially Theorems 11.1 and 11.7 and Theorem 12.3, p. 90. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Chapter 7, Propositions 7.1, 7.4 and 7.5. Nakahara, §7.3 and §7.8.